

MATH3092: Final Project

Conformal Prediction Under Class Imbalance and Temporal Shift

A tabular classification study using WDBC and chronological Bank Marketing
data

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Abstract

Conformal prediction turns a point prediction into a prediction set, so it can report uncertainty rather than forcing every observation into one class label. Its standard finite-sample coverage guarantee, however, is not automatic in every deployment setting. The guarantee relies on exchangeability between calibration scores and future test scores. This report studies two common departures from this condition in binary tabular classification: class imbalance and temporal shift.

The empirical design has three goals. First, the Wisconsin Diagnostic Breast Cancer dataset (WDBC) is used as a near-exchangeable benchmark, because the train–calibration–test split is stratified and random. Second, a controlled synthetic-shift experiment on WDBC tests whether larger calibration–test score mismatch leads to larger coverage loss. Third, the Bank Marketing data from the UCI Machine Learning Repository are evaluated chronologically, so that the calibration block always precedes the test block in time. I use `bank-full.csv` as the main Bank experiment and `bank-additional-full.csv` only as a robustness check with a different feature set.

At the pre-specified level $\alpha = 0.10$, WDBC behaves as expected: Split Conformal Prediction (SplitCP), Mondrian SplitCP, and smoothed Adaptive Prediction Sets (APS) remain close to the nominal 0.90 coverage target. Under synthetic WDBC shift, coverage falls as the calibration–test score Wasserstein distance increases. The chronological Bank experiment shows the practical consequence of calibration–test mismatch: in a 3×10 robustness grid on `bank-full.csv`, vanilla SplitCP never reaches the 0.90 target, and its failure is most severe for the positive class. Mondrian calibration reduces this class-wise failure more cleanly than the classifier-weighted baseline, although it does not fully restore nominal coverage. The weighted method is therefore treated as a retrospective, batch-level adjustment that uses test-block covariates, not as a general guarantee under temporal shift. I also include a small perturbation argument showing that the coverage distortion caused by weight clipping alone is bounded by twice the clipping total variation; this argument does not establish validity of the weighted baseline under temporal shift, but it separates clipping distortion from other sources of mismatch. Overall, the report shows that conformal validity should be checked under the intended calibration–deployment relationship rather than assumed from exchangeability in the abstract.

Acronyms

APS	Adaptive Prediction Sets
AUC	Area under the receiver operating characteristic curve
ESS	Effective sample size
HGB	HistGradientBoosting
KS	Kolmogorov–Smirnov
LR	Logistic Regression
SplitCP	Split Conformal Prediction
TV	Total variation
TWM	Temporally weighted Mondrian; used only in the exploratory appendix
WDBC	Wisconsin Diagnostic Breast Cancer dataset

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1 Introduction

Binary classifiers usually return one predicted class label. That is convenient, but it can obscure uncertainty. In a diagnosis, credit, or marketing-response problem, a confidently incorrect prediction can be more damaging than an output that admits ambiguity. Conformal prediction addresses this by returning a set of plausible labels. In a binary problem the set may be $\{0\}$, $\{1\}$, $\{0, 1\}$, or, for procedures that allow empty sets, $\emptyset$.

The standard appeal of conformal prediction is its finite-sample marginal coverage guarantee. If the calibration scores and the future test score are exchangeable, then a split conformal prediction set covers the true label with probability at least $1 - \alpha$. The guarantee is conditional: it holds only when calibration and test scores are exchangeable, and it is distribution-free under that assumption rather than a guarantee for arbitrary temporal deployment. This report focuses on two reasons why the assumption can become fragile in tabular binary classification. First, class imbalance can hide poor minority-class coverage behind a reasonable marginal average. Second, temporal shift can make the calibration score distribution different from the later test score distribution.

The report is organised around three questions.

1. In a benchmark setting where exchangeability is more plausible, do the conformal procedures behave as expected?
2. If calibration and test scores are deliberately separated by a controlled shift, does larger score-distribution mismatch correspond to larger coverage loss?
3. In chronological Bank Marketing data, where calibration data precede deployment data in time, which procedures still give useful coverage and which failures become visible only through class-wise and rolling-window diagnostics?

The datasets are used for different purposes, and this distinction is important. WDBC is the positive-control benchmark: stratified random splits make the conformal assumption more plausible, so this dataset checks that the pipeline itself is working. The synthetic-shift WDBC experiment is a bridge between the benchmark and the deployment setting: it keeps the data source fixed while increasing calibration–test mismatch in a controlled way. The main deployment-style experiment is the chronological Bank Marketing dataset from the UCI Machine Learning Repository, specifically `bank-full.csv`. I call this dataset *Original Bank* throughout the report. The dataset `bank-additional-full.csv` is called *Additional Bank*; it is used only as a robustness check because it has a different feature set and weaker base-model discrimination in the without-duration setting.

The methods also have different roles. SplitCP is the vanilla marginal-coverage baseline. Mondrian SplitCP calibrates separately within each class and is therefore aimed at class imbalance. APS changes the conformity score and can improve efficiency when the base probabilities are informative. The classifier-weighted SplitCP baseline is a practical covariate-shift adjustment; because it uses unlabelled covariates from the later test block, it should be read as a retrospective, batch-level adjustment, not as a strictly online deployment method.

Contributions. The report makes three contributions. First, it compares SplitCP, Mondrian SplitCP, APS, and classifier-weighted SplitCP across a near-exchangeable benchmark, a controlled synthetic-shift experiment, and chronological Bank Marketing evaluations. Second, it links empirical under-coverage to calibration–test score drift using Wasserstein and Kolmogorov–Smirnov diagnostics. Third, it includes a small technical clarification for the weighted baseline: an elementary perturbation argument shows that the coverage distortion caused by weight clipping alone is controlled by twice the clipping total variation, $2TV_{\text{clip}}$. This perturbation argument is deliberately limited in scope. It separates clipping distortion from density-ratio estimation error and from deeper violations such as label shift or concept drift; it does not prove that the implemented weighted method is valid under temporal shift.

2 Theoretical background

This section introduces the basic conformal terminology before the empirical sections begin. The main point is simple: conformal prediction gives a finite-sample guarantee under exchangeability, but the guarantee is only as relevant as that assumption is to the deployment setting.

2.1 Prediction sets and marginal validity

Let $\mathcal{Y} = \{0, 1\}$ be the label space. For a feature vector x , a conformal classifier returns a prediction set $\hat{\Gamma}_\alpha(x) \subseteq \mathcal{Y}$ rather than a single class label. The nominal miscoverage level is α , so the target coverage level is $1 - \alpha$. The usual marginal-validity statement is

$$\mathbb{P}(Y_{n+1} \in \hat{\Gamma}_\alpha(X_{n+1})) \geq 1 - \alpha. \quad (1)$$

Here the probability is averaged over future test observations. It is *marginal*, not conditional on a particular feature value $X = x$. This distinction matters for imbalanced data: a method may reach the target average coverage while still covering one class much less often than the other (Shafer and Vovk, 2008; Vovk et al., 2005; Angelopoulos and Bates, 2021).

2.2 Exchangeability and finite-sample calibration

Exchangeability means that the joint distribution of the labelled observations is unchanged by permuting their order. Independent and identically distributed sampling is one common way to obtain exchangeability, but it is a stricter condition than exchangeability requires. Split conformal calibration uses exchangeability because, under this assumption, the future test score has a uniform rank among the calibration scores plus the test score itself. This rank argument gives the finite-sample coverage guarantee. If the future test score is drawn from a different distribution than the calibration scores, the rank argument no longer applies directly.

2.3 Class imbalance and class-conditional validity

Class imbalance creates a reporting problem. Suppose the negative class is common and easy to cover, while the positive class is rare and poorly covered. The marginal average can still look acceptable because the majority class dominates the average. Mondrian conformal prediction is designed to address this issue by calibrating a separate threshold for each class. Under within-class exchangeability, it gives class-conditional coverage. In practice, it can improve minority-class coverage, but it may also increase prediction-set size and it does not automatically solve temporal shift.

2.4 Why temporal shift breaks conformal validity

Temporal shift means that the distribution of later deployment observations differs from the distribution of earlier calibration observations. The difference may come from covariate shift, label shift, concept drift, or a mixture of these. For conformal prediction, the immediate object is the score distribution: if true-label nonconformity scores in the test block are not exchangeable with calibration scores, the calibrated threshold may be too small and under-coverage can result. Therefore, chronological evaluation is not a cosmetic choice in this report; it is the direct test of the exchangeability assumption underlying the conformal coverage guarantee.

2.5 Practical versus formal validity

Some modifications make prediction sets easier to use but change the formal conformal object. APS(+fallback) is the clearest example in this report. The raw APS method may return an empty set. The fallback version replaces an empty set by the singleton class with the highest predicted probability. That replacement is operationally convenient, but it is a post-processing rule. Its empirical coverage is still useful to report, but it should not be interpreted in the same way as the coverage of the unmodified conformal set.

2.6 Related work

Finite-sample validity and its limits. Classical conformal prediction establishes finite-sample marginal coverage under exchangeability (Shafer and Vovk, 2008; Vovk et al., 2005; Angelopoulos and Bates, 2021). That result motivates the WDBC baseline, where the random stratified split makes exchangeability more plausible. The same literature also explains why this guarantee should not be read as unconditional deployment validity. Barber et al. (2023) study conformal prediction beyond exchangeability and show that coverage degradation depends on how different the calibration and test score distributions are. This directly motivates the score-drift diagnostics used later in the WDBC synthetic-shift experiment and the Bank experiments.

Class-conditional and adaptive set construction. Romano et al. (2020) introduce Adaptive Prediction Sets (APS), which add labels in decreasing order of predicted probability until a calibrated

cumulative-probability threshold is reached. APS can give smaller sets for observations whose class probabilities are decisive and larger sets for ambiguous observations. In this report, APS improves efficiency on WDBC, but it does not remove the main Original Bank failure. On the chronological Original Bank data, APS(allow-empty) stays close to SplitCP, which suggests that the dominant problem is not only the choice of score but the mismatch between calibration and test score distributions.

Conformal prediction under distribution shift. Tibshirani et al. (2019) propose weighted conformal prediction for covariate shift. The ideal version uses known density ratios to reweight calibration scores toward the test covariate distribution. In practice the ratios must be estimated, and the guarantee becomes assumption-dependent. The classifier-weighted baseline in this report follows the same idea at a practical level: a domain classifier estimates which calibration observations look most similar to the later test block. Because the procedure uses unlabelled future test covariates, it is best interpreted as a retrospective, batch-level adjustment rather than a method that could be deployed sequentially, one observation at a time, without future information.

Online and adaptive conformal inference. Another literature studies conformal inference in fully online or time-adaptive settings. Gibbs and Candès (2021) and Zaffran et al. (2022) update thresholds over time as new observations arrive. These methods are relevant to temporal deployment, but they operate under a different protocol from this report. My Bank experiments use fixed chronological splits and rolling windows, not sequential feedback after every test point. Therefore, the comparison is conceptual: online adaptive methods are natural future work, but they are not direct baselines for the experiments reported here.

The main report does not propose a validated new conformal algorithm. Its contribution is an empirical study of standard conformal procedures and one practical weighting repair when the evaluation moves from a near-exchangeable benchmark to chronologically ordered tabular data; the exploratory TWM appendix is included only as a mechanism check.

3 Methods

This section describes the base classifiers, the nonconformity score, and the four set-construction procedures compared in this report. Throughout, a larger nonconformity score means that a candidate label is less compatible with the fitted model’s predicted probability.

3.1 Base probabilistic models

Two base models are used throughout: Logistic Regression (LR) and HistGradientBoosting (HGB). LR is a simple linear baseline whose probabilities are often comparatively stable. HGB is a stronger non-linear tabular model, but boosting-based probabilities can still be miscalibrated even when ranking performance is good. Using both models checks whether a conformal failure is tied to one probability model or appears across different model classes. When the results report AUC, that AUC is computed from the base-model probabilities before conformal prediction is applied.

3.2 Nonconformity score

For a feature vector x and candidate class label $y \in \{0, 1\}$, let $\hat{p}(y | x)$ be the fitted model’s predicted probability for label y . I use the probability-based nonconformity score

$$s(x, y) = 1 - \hat{p}(y | x). \quad (2)$$

A label with high predicted probability has a small score and is more likely to be included in the prediction set. A label with low predicted probability has a large score and is included only if the conformal threshold is large enough.

3.3 Split Conformal Prediction

SplitCP uses a separate calibration set to choose one global score threshold. Let I_{cal} be the calibration index set and let $S_i = s(X_i, Y_i)$ be the true-label score for calibration point i . If $m = |I_{\text{cal}}|$, define

$$k = \lceil (m + 1)(1 - \alpha) \rceil, \quad \hat{q}_{1-\alpha} = S_{(k)}, \quad (3)$$

where $S_{(k)}$ is the k th smallest calibration score. For a new observation x , SplitCP includes every label whose score is no larger than this threshold:

$$\hat{\Gamma}_\alpha(x) = \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}_{1-\alpha}\}. \quad (4)$$

In words, a candidate label is included if the fitted model does not find it more unusual than the calibrated $(1 - \alpha)$ cutoff.

Proposition 1 (Finite-sample marginal coverage). *Conditional on the fitted model, if the m calibration scores and the future test score are exchangeable, then*

$$\mathbb{P}(Y_{n+1} \in \hat{\Gamma}_\alpha(X_{n+1})) \geq 1 - \alpha.$$

Proof sketch. Let $S_1, \dots, S_m$ be the calibration scores and let $S_{m+1} = s(X_{n+1}, Y_{n+1})$ be the future true-label test score. Under exchangeability, the vector $(S_1, \dots, S_m, S_{m+1})$ is permutation-symmetric. Therefore the rank of S_{m+1} among these $m + 1$ scores is uniform on $\{1, \dots, m + 1\}$. With $k = \lceil (m + 1)(1 - \alpha) \rceil$, the event $S_{m+1} \leq S_{(k)}$ has probability at least $1 - \alpha$. By Equation (4), this event is exactly

$$Y_{n+1} \in \hat{\Gamma}_\alpha(X_{n+1}) \iff s(X_{n+1}, Y_{n+1}) \leq \hat{q}_{1-\alpha}.$$

This proves the marginal coverage statement. The same proof also shows where temporal deployment can fail: if the future score is no longer exchangeable with the calibration scores, the rank argument is no longer valid.

3.4 Mondrian SplitCP

Mondrian SplitCP uses a separate threshold for each class. Let m_y be the number of calibration points with label y , and let $S_{(k_y)}^{(y)}$ be the k_y th order statistic among calibration scores from class y . Define

$$k_y = \lceil (m_y + 1)(1 - \alpha) \rceil, \quad \hat{q}_{1-\alpha}(y) = S_{(k_y)}^{(y)}, \quad \hat{\Gamma}_\alpha(x) = \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}_{1-\alpha}(y)\}. \quad (5)$$

Under exchangeability within each class stratum, Mondrian gives finite-sample class-conditional coverage. This is why it is the main class-imbalance repair in the report.

3.5 Smoothed APS

APS uses the whole probability vector rather than checking each candidate label against a fixed score independently (Romano et al., 2020). Let π order the labels by decreasing fitted probability. For the label in position j , the smoothed APS score is

$$s_{\text{APS}}(x, \pi(j)) = \sum_{\ell < j} \hat{p}(\pi(\ell) \mid x) + U \hat{p}(\pi(j) \mid x), \quad U \sim \text{Unif}(0, 1). \quad (6)$$

The random variable U is the smoothing term used to avoid deterministic boundary effects. I report APS in two modes. *APS (allow-empty)* is the raw conformal version and may return $\emptyset$. *APS (+fallback)* replaces an empty raw set by the top-probability singleton. The fallback version is useful operationally, but because it changes the set after conformal calibration, I treat it as a non-conformal post-processing baseline.

3.6 Weighted SplitCP

The Bank experiments include a classifier-weighted SplitCP baseline motivated by weighted conformal prediction under covariate shift (Tibshirani et al., 2019). The aim is not to restore a distribution-free theorem under arbitrary temporal shift. The narrower aim is to give more influence to calibration points whose covariates look similar to the later test block.

In the implementation used for the main Bank experiments, a Logistic Regression domain classifier distinguishes an earlier source block, consisting of training and calibration covariates, from a later target block, consisting of test covariates. The domain-classifier probabilities are clipped to avoid extreme odds and then converted into prior-corrected, density-ratio-like weights. The resulting threshold is test-point-specific. Let S_i denote calibration scores and let $\hat{w}_i$ denote estimated calibration weights. For a test point x with estimated weight $\hat{w}(x)$, define

$$\hat{q}_{1-\alpha}^w(x) = \inf \left\{ q : \frac{\sum_{i \in I_{\text{cal}}} \hat{w}_i \mathbf{1}\{S_i \leq q\}}{\sum_{i \in I_{\text{cal}}} \hat{w}_i + \hat{w}(x)} \geq 1 - \alpha \right\}. \quad (7)$$

The weighted prediction set is

$$\hat{\Gamma}_\alpha^w(x) = \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}_{1-\alpha}^w(x)\}. \quad (8)$$

Two caveats are essential. First, the method mainly targets covariate-shift-like mismatch. It can still fail under label shift or concept drift. Second, because it uses unlabelled covariates from the later test block, it is a retrospective, batch-level adjustment rather than a strictly online deployment procedure.

4 Evaluation metrics

The evaluation considers three questions: whether the true label is covered on average, whether coverage is balanced across classes, and how large the returned prediction sets are. For test observations (X_i, Y_i) with prediction sets $\Gamma_i = \hat{\Gamma}_\alpha(X_i)$ and n_{test} total test points, the primary metrics are:

$$\hat{C} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbf{1}\{Y_i \in \Gamma_i\}, \quad (\text{marginal coverage}) \quad (9)$$

$$\hat{C}_y = \frac{1}{n_y} \sum_{i: Y_i=y} \mathbf{1}\{Y_i \in \Gamma_i\}, \quad (\text{per-class coverage}) \quad (10)$$

$$\hat{E} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |\Gamma_i|. \quad (\text{average set size}) \quad (11)$$

Smaller $\hat{E}$ is better when comparing methods with similar coverage levels.

Two derived summaries are used repeatedly in the Bank tables:

$$\hat{G} = \hat{C} - (1 - \alpha), \quad (\text{coverage gap relative to target}) \quad (12)$$

$$\hat{D} = |\hat{C}_1 - \hat{C}_0|. \quad (\text{class-disparity gap}) \quad (13)$$

Thus $\hat{G} < 0$ indicates under-coverage and $\hat{D}$ summarises class asymmetry.

For allow-empty conformal methods, I also report the empty-set rate

$$\frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbf{1}\{\Gamma_i = \emptyset\}.$$

For APS(+fallback), I report the fallback-trigger rate, namely the raw APS empty rate before replacement by a singleton.

For rolling-window diagnostics, let the test period be partitioned into windows $w = 1, \dots, W$. The local coverage in window w is

$$\hat{C}^{(w)} = \frac{1}{n_w} \sum_{i \in w} \mathbf{1}\{Y_i \in \Gamma_i\}. \quad (14)$$

The rolling-window summary tables report the mean, minimum, and maximum of these local coverages across windows, together with the mean local coverage gap.

4.1 Mechanism diagnostics: score drift and weighting distortion

Coverage summaries say that a method failed; they do not by themselves say why it failed. I therefore record two diagnostics that connect the empirical results to calibration-test mismatch. First, for the true-label nonconformity scores $S_i = 1 - \hat{p}(Y_i | X_i)$, I compare the empirical calibration and test score distributions using the Kolmogorov–Smirnov distance

$$D_{\text{KS}}(\hat{F}_{\text{cal}}, \hat{F}_{\text{test}}) = \sup_t |\hat{F}_{\text{cal}}(t) - \hat{F}_{\text{test}}(t)| \quad (15)$$

and the 1-Wasserstein distance

$$W_1(\hat{F}_{\text{cal}}, \hat{F}_{\text{test}}) = \int_{\mathbb{R}} |\hat{F}_{\text{cal}}(t) - \hat{F}_{\text{test}}(t)| dt. \quad (16)$$

These are used as *proxy metrics* for score-distribution mismatch, both globally and within class. They are not the theorem objects in Barber et al. (2023), so they should be read as mechanism checks rather than as direct finite-sample degradation bounds.

Second, for the classifier-weighted baseline I report the calibration effective sample size

$$\text{ESS}(w) = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} \quad (17)$$

and a clipping-distortion diagnostic. If w_i^{raw} denotes the unnormalised domain-classifier odds and w_i^{clip} the clipped version used in the experiment, define the normalised weights $\tilde{w}_i^{\text{raw}} = w_i^{\text{raw}} / \sum_j w_j^{\text{raw}}$ and $\tilde{w}_i^{\text{clip}} = w_i^{\text{clip}} / \sum_j w_j^{\text{clip}}$, and report

$$\text{TV}_{\text{clip}} = \frac{1}{2} \sum_i |\tilde{w}_i^{\text{raw}} - \tilde{w}_i^{\text{clip}}|. \quad (18)$$

Here ESS is a stability diagnostic for the weighted quantile, while TV_{clip} measures how far the implemented weighting rule moves away from the unclipped odds distribution. Neither quantity is itself a coverage bound; the point is to separate weight instability from target-distribution distortion.

4.2 A perturbation lemma for clipped weighted calibration

The next lemma has a narrow purpose. It does not prove that classifier-weighted SplitCP is valid under temporal shift. It only quantifies how much coverage can be lost from the clipping step, assuming ideal weighted exchangeability already holds. The weighted-conformal premise follows Tibshirani et al. (2019). The perturbation bound below is an elementary derivation for the clipping step used in this implementation; it is not a new conformal algorithm or a new weighted-conformal validity theorem. Assume also that the weighted quantiles are well-defined at the target level, using the standard weighted-conformal convention for the test-point self-weight, or an equivalent conservative implementation rule when the calibration mass is insufficient. Fix a test point x and calibration scores $S_1, \dots, S_m \in \mathbb{R}$. Let $\tilde{w} = (\tilde{w}_1, \dots, \tilde{w}_m, \tilde{w}_{m+1})$ denote ideal normalised weights, where $\tilde{w}_{m+1}$ is the self-weight of the test point, and let $\tilde{w}^{\text{clip}} = (\tilde{w}_1^{\text{clip}}, \dots, \tilde{w}_{m+1}^{\text{clip}})$ denote the implemented clipped version. Define

$$F(t) = \sum_{i=1}^m \tilde{w}_i \mathbf{1}\{S_i \leq t\}, \quad F^{\text{clip}}(t) = \sum_{i=1}^m \tilde{w}_i^{\text{clip}} \mathbf{1}\{S_i \leq t\},$$

and let

$$\varepsilon := \text{TV}_{\text{clip}} = \frac{1}{2} \sum_{i=1}^{m+1} |\tilde{w}_i - \tilde{w}_i^{\text{clip}}|.$$

For level $1 - \alpha$, write

$$\hat{q}_{1-\alpha}(x) = \inf\{t : F(t) \geq 1 - \alpha\}, \quad \hat{q}_{1-\alpha}^{\text{clip}}(x) = \inf\{t : F^{\text{clip}}(t) \geq 1 - \alpha\},$$

and let the corresponding prediction sets be $\hat{\Gamma}_\alpha$ and $\hat{\Gamma}_\alpha^{\text{clip}}$. Assume the ideal weights satisfy the weighted-exchangeability conditions of Tibshirani et al. (2019), so that

$$\mathbb{P}(Y_{n+1} \in \hat{\Gamma}_\alpha(X_{n+1})) \geq 1 - \alpha.$$

Lemma 1 (Clipping-induced coverage perturbation). *Under the setup above,*

$$\mathbb{P}(Y_{n+1} \in \hat{\Gamma}_\alpha^{\text{clip}}(X_{n+1})) \geq 1 - \alpha - 2\varepsilon.$$

Equivalently, clipping-induced coverage deviation is linearly controlled by twice the normalised clipping total variation.

Proof. For any t ,

$$|F(t) - F^{\text{clip}}(t)| = \left| \sum_{i=1}^m (\tilde{w}_i - \tilde{w}_i^{\text{clip}}) \mathbf{1}\{S_i \leq t\} \right| \leq \sum_{i=1}^m |\tilde{w}_i - \tilde{w}_i^{\text{clip}}| \leq 2\varepsilon.$$

Hence $\sup_t |F(t) - F^{\text{clip}}(t)| \leq 2\varepsilon$. Let $q^* = \hat{q}_{1-\alpha}^{\text{clip}}(x)$. Since $F^{\text{clip}}(q^*) \geq 1 - \alpha$, the uniform bound gives

$$F(q^*) \geq F^{\text{clip}}(q^*) - 2\varepsilon \geq 1 - \alpha - 2\varepsilon.$$

Now set $\alpha' = \alpha + 2\varepsilon$. By definition of the ideal weighted quantile, the previous display implies

$$\hat{q}_{1-\alpha}^{\text{clip}}(x) \geq \hat{q}_{1-\alpha'}(x),$$

so the clipped prediction set contains the ideal set at the relaxed level:

$$\hat{\Gamma}_{\alpha}^{\text{clip}}(x) \supseteq \hat{\Gamma}_{\alpha'}(x).$$

Applying weighted exchangeability at level $1 - \alpha'$ yields

$$\mathbb{P}(Y_{n+1} \in \hat{\Gamma}_{\alpha}^{\text{clip}}(X_{n+1})) \geq \mathbb{P}(Y_{n+1} \in \hat{\Gamma}_{\alpha'}(X_{n+1})) \geq 1 - \alpha' = 1 - \alpha - 2\varepsilon.$$

□

Remark. Lemma 1 separates clipping distortion from two other sources of error: density-ratio estimation error in the domain classifier and failure of weighted exchangeability under label shift or concept drift. It therefore does *not* establish nominal coverage for the implemented weighted baseline. What it does show is that clipping distortion is one identifiable source of error. Because the bound is conservative, it should not be used to attribute the observed coverage gaps precisely; density-ratio estimation error and deeper distributional mismatch remain possible explanations.

5 Datasets and experimental design

This section clarifies the role of each dataset before the results are reported. The three datasets are not interchangeable: WDBC checks the pipeline under a more favourable random-split setting; Original Bank is the main chronological deployment test; Additional Bank is a secondary robustness check with a different feature set.

WDBC benchmark. The Wisconsin Diagnostic Breast Cancer dataset (WDBC) is used as the near-exchangeable benchmark. It contains 569 observations, 30 real-valued features, and binary labels M (malignant) and B (benign); malignant is mapped to label 1 and treated as the minority class, while benign is mapped to label 0 (Wolberg et al., 1993). The WDBC experiments use stratified random 60/20/20 train–calibration–test splits over seeds 1–30, and preprocessing is fit on the training split only. Because the splits are random and stratified, WDBC is the setting where the exchangeability assumption is most plausible in this report.

Bank Marketing datasets. The main chronological stress test uses the Bank Marketing dataset (`bank-full.csv`) from the UCI Machine Learning Repository (UCI Machine Learning Repository, 2012), which I call *Original Bank*. It contains 45,211 rows and a positive response rate of 11.7% (Moro et al., 2014). The secondary robustness check uses `bank-additional-full.csv`, which I call *Additional Bank*. It contains 41,188 rows and a positive response rate of 11.3%. Additional Bank is not treated as a second main leaderboard because it has a different feature set and different base-model discrimination at the same time.

Both Bank studies use LR and HGB, miscoverage levels $\alpha \in \{0.05, 0.10, 0.20\}$, single chronological splits with and without `duration`, and rolling-window evaluation without `duration` in the main analysis. The main text focuses on $\alpha = 0.10$, a middle operating point where failures are visible without using the most aggressive target. The appendices report results for $\alpha = 0.05$ and 0.20 so that the qualitative pattern can be checked across levels.

Chronological splitting and leakage control. The key Bank design choice is chronological train $\rightarrow$ calibration $\rightarrow$ test splitting. In the single-split experiments, the earliest 60% of observations are used for training, the next 20% for calibration, and the final 20% for testing. No shuffling is used. The variable `duration` is excluded from the substantive analysis because it is a known leakage variable: it is only observed after the phone call has taken place. I include `duration` only in a diagnostic ablation to show that better ranking performance does not automatically fix conformal reliability.

Rolling windows and robustness grid. The rolling-window diagnostic checks whether coverage collapses locally in time. The core rolling design uses overlapping chronological windows with train/calibration/test fractions 0.50/0.20/0.10 and step size 0.05, giving five overlapping windows on the Bank datasets. The overlap is deliberate: it gives a coarse time path of reliability rather than five unrelated mini-experiments.

To check that the headline Original Bank result is not an artefact of one split or one HGB seed, the $\alpha = 0.10$ without-duration experiment is also repeated on a 3×10 robustness grid: `train_frac` $\in \{0.55, 0.60, 0.65\}$ crossed with 10 random seeds. This gives 30 runs per model–method cell. The

resulting mean $\pm$ standard deviation values are reported in Table 4. The appendix keeps the single-seed cross- α tables as supporting records rather than headline evidence.

Weighted baseline protocol. For the classifier-weighted baseline, the domain classifier distinguishes an earlier source block from the later test block, clips the resulting probabilities to avoid extreme odds, and converts the clipped probabilities into density-ratio-like weights. Because this procedure uses unlabelled covariates from the later test block, it should be read as a retrospective batch adjustment. It is useful for asking whether covariate-shift-style weighting could help, but it is not a strict online deployment method.

Table 1: Deployment-oriented Bank Marketing protocol.

Component	Design choice	Purpose
Main split	Chronological train $\rightarrow$ calibration $\rightarrow$ test	Preserve deployment order and directly test exchangeability failure
Leakage control	Exclude duration in main analysis; include only in ablation	Separate realistic evaluation from leakage-driven optimism
Temporal diagnostics	Rolling-window recalibration and testing	Measure local coverage over time rather than only a global average
Shift-aware baseline	Classifier-based weighting	Target covariate-shift-like mismatch with a practical baseline
Primary summaries	Coverage gap, class-disparity gap, average set size	Distinguish overall validity from class-wise reliability and efficiency

6 Results I: WDBC baseline under approximate exchangeability

This first result section is a sanity check rather than the main deployment claim. WDBC uses stratified random splits, so calibration and test data are much closer to the setting required by the split conformal theorem than the chronological Bank experiments. In this favourable setting, the conformal pipeline behaves much as expected. The base models are very strong, with test AUC 0.994 for LR and 0.991 for HGB, and empirical coverage stays close to the nominal target across α values. That is why WDBC is treated as a benchmark rather than the main result. The discussion in the main text focuses on the pre-specified operating point $\alpha = 0.10$, while the appendices keep $\alpha = 0.05$ and 0.20 visible to show that the qualitative pattern is not tied to that single threshold.

Table 2: Base classifier AUC on WDBC (test split). Mean $\pm$ sd across 30 stratified splits.

Base model	Test AUC (mean $\pm$ sd)
LR	0.994 $\pm$ 0.007
HGB	0.991 $\pm$ 0.011

Table 3: WDBC results at $\alpha = 0.10$. APS(+fallback) is reported as a non-conformal operational baseline.

Model	Method	Cov	Cov(1)	Cov(0)	Avg size	Empty/fallback
LR	SplitCP	0.893	0.878	0.902	0.900	0.100
LR	Mondrian	0.892	0.909	0.881	0.898	0.102
LR	APS (allow-empty)	0.900	0.890	0.906	0.975	0.089
LR	APS (+fallback)	0.988	0.976	0.996	1.064	0.089
HGB	SplitCP	0.890	0.870	0.903	0.901	0.099
HGB	Mondrian	0.892	0.914	0.878	0.909	0.091
HGB	APS (allow-empty)	0.897	0.881	0.906	0.946	0.078
HGB	APS (+fallback)	0.974	0.957	0.984	1.024	0.078

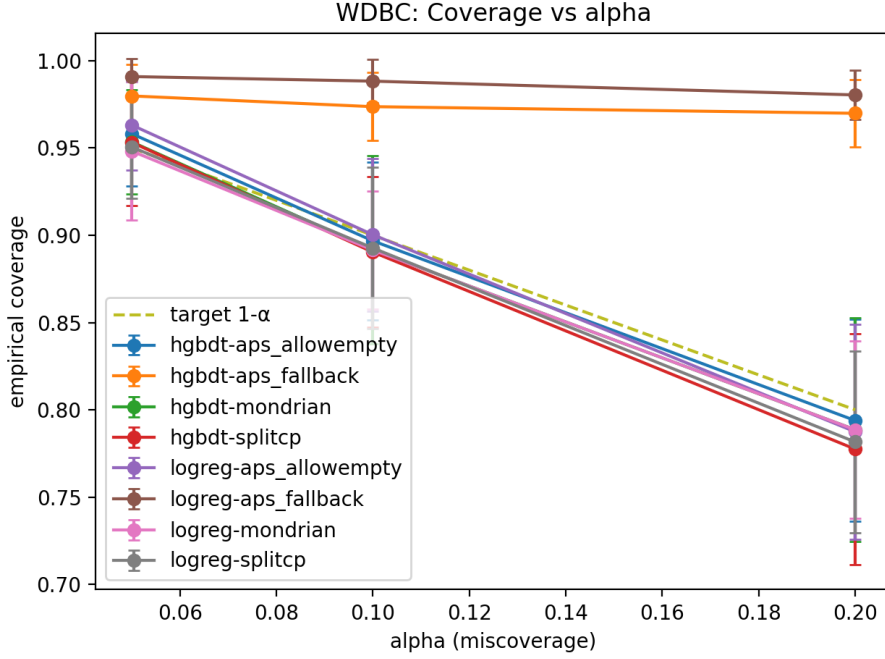
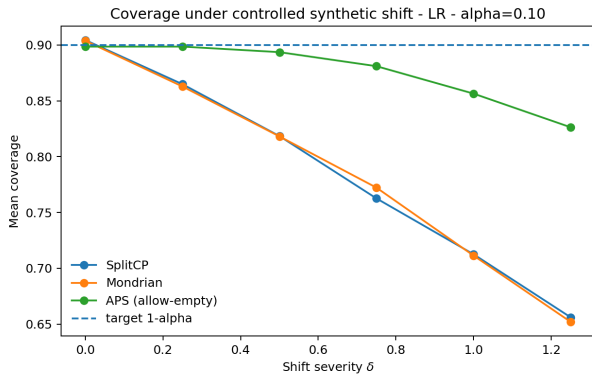


Figure 1: WDBC marginal coverage versus α . The dashed line marks the target $1 - \alpha$. Under approximate exchangeability, the conformal procedures stay close to nominal coverage and the pipeline behaves as expected.

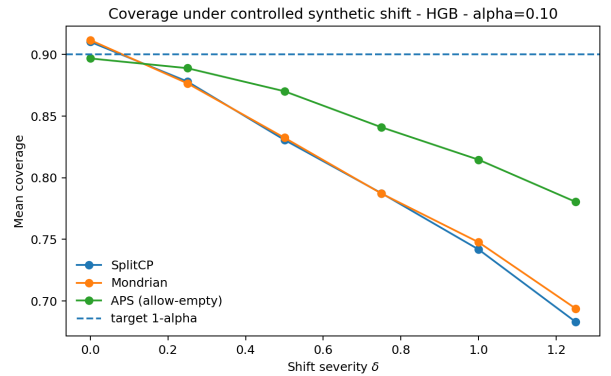
The WDBC result should be read narrowly. It shows that the implementation behaves sensibly in a setting where the core theorem is plausible. It does not show that conformal prediction stays reliable once that setting changes. Mondrian reallocates some protection across classes, and APS(+fallback) remains an operational baseline rather than a formal conformal guarantee.

6.1 WDBC as a controlled bridge experiment under synthetic shift

The second WDBC experiment explains why the later Bank failure is plausible. I perturb only the WDBC test block along a discriminative direction estimated from the training data, while leaving the training and calibration blocks unchanged. The perturbation strength is indexed by δ . I do not use this experiment to mimic the Bank datasets feature by feature. Instead, I use it to test whether larger calibration–test score mismatch is systematically accompanied by larger negative coverage gaps under a clean, controlled shift.



(a) LR, coverage under increasing synthetic shift.



(b) HGB, coverage under increasing synthetic shift.

Figure 2: Controlled WDBC bridge experiment at $\alpha = 0.10$. As shift severity δ increases, SplitCP and Mondrian degrade almost monotonically; APS(allow-empty) decays more slowly but is not immune. Key score-drift check: the Pearson correlation between global score Wasserstein distance and coverage gap is about -0.934 for LR SplitCP and -0.883 for HGB SplitCP.

The synthetic-shift results show a clear loss of coverage as mismatch grows. For LR SplitCP, mean coverage falls from 0.904 at $\delta = 0$ to 0.656 at $\delta = 1.25$; for HGB SplitCP it falls from 0.910 to 0.683. Mondrian follows the same pattern. APS(allow-empty) is somewhat more resilient at moderate shift levels, but it also deteriorates, dropping from 0.899 to 0.826 for LR and from 0.897 to 0.780 for HGB. Over the same range, the global score-Wasserstein proxy rises from 0.020 to 0.164 for LR

and from 0.019 to 0.154 for HGB. The relationship is strong: the Pearson correlation between global Wasserstein distance and coverage gap is about -0.934 for LR SplitCP and -0.883 for HGB SplitCP. This matters because it gives the later Bank results a clearer mechanism. As the calibration and test score distributions separate, under-coverage rises in a systematic way.

7 Results II: Original Bank under a single chronological split

Original Bank is the main deployment-style experiment. Here the calibration block is earlier in time than the test block, so the evaluation directly tests whether conformal calibration transfers forward in time. The dataset is much larger than WDBC, but that does not protect the conformal guarantee. Because the observations stay in time order, calibration–test mismatch is exposed rather than averaged away. The main results therefore focus on the without-duration setting.

Table 4: Original Bank (`bank-full.csv`), $\alpha = 0.10$, without `duration`. Mean $\pm$ std over 3 chronological cut-points (`train_frac` $\in \{0.55, 0.60, 0.65\}$) $\times$ 10 random seeds = 30 runs per cell. Supersedes the single-seed version of the earlier draft.

Model	Method	Cov	Cov(1)	Cov(0)	Avg size	Cov. gap	Dispar.
LR	SplitCP	0.721 $\pm$ 0.027	0.147 $\pm$ 0.048	1.000 $\pm$ 0.000	1.109 $\pm$ 0.045	−0.179 $\pm$ 0.027	0.853 $\pm$ 0.048
LR	Mondrian	0.876 $\pm$ 0.016	0.794 $\pm$ 0.127	0.906 $\pm$ 0.034	1.570 $\pm$ 0.127	−0.024 $\pm$ 0.016	0.152 $\pm$ 0.120
LR	APS (allow-empty)	0.721 $\pm$ 0.009	0.214 $\pm$ 0.076	0.966 $\pm$ 0.016	1.125 $\pm$ 0.090	−0.179 $\pm$ 0.009	0.752 $\pm$ 0.061
LR	Weighted SplitCP	0.788 $\pm$ 0.014	0.341 $\pm$ 0.122	1.000 $\pm$ 0.000	1.242 $\pm$ 0.112	−0.112 $\pm$ 0.014	0.659 $\pm$ 0.122
HGB	SplitCP	0.755 $\pm$ 0.012	0.243 $\pm$ 0.106	1.000 $\pm$ 0.000	1.191 $\pm$ 0.101	−0.145 $\pm$ 0.012	0.757 $\pm$ 0.106
HGB	Mondrian	0.852 $\pm$ 0.023	0.875 $\pm$ 0.080	0.833 $\pm$ 0.056	1.606 $\pm$ 0.083	−0.048 $\pm$ 0.023	0.121 $\pm$ 0.045
HGB	APS (allow-empty)	0.758 $\pm$ 0.011	0.279 $\pm$ 0.086	0.988 $\pm$ 0.007	1.212 $\pm$ 0.098	−0.142 $\pm$ 0.011	0.709 $\pm$ 0.079
HGB	Weighted SplitCP	0.854 $\pm$ 0.029	0.538 $\pm$ 0.149	1.000 $\pm$ 0.000	1.439 $\pm$ 0.160	−0.046 $\pm$ 0.029	0.462 $\pm$ 0.149

Note. Entries such as $\text{Cov}(0) = 1.000 \pm 0.000$ for vanilla and weighted SplitCP are not numerical precision artefacts. In those runs, the learned thresholds made label 0 effectively always included on negative cases, so negative-class coverage was deterministically one and its reported standard deviation is exactly zero.

The pattern is consistent across cut-points and seed settings. **Across all 30 robustness-grid cells, vanilla SplitCP never reached the nominal 0.90 target: LR SplitCP coverage averaged 0.721 and HGB SplitCP 0.755.** This should be read as a stable empirical pattern across the scanned chronological cut-points and seed settings, rather than as a one-split anomaly. The failure is also strongly class-asymmetric. For LR, positive-class coverage averages $\hat{C}_1 = 0.147 \pm 0.048$, while negative-class coverage is exactly 1.000 in every run. HGB is less extreme, but still badly imbalanced, with $\hat{C}_1 = 0.243 \pm 0.106$.

Mondrian improves that pattern more cleanly than the other alternatives in this scan. Across 30 runs, marginal coverage rises to 0.876 ± 0.016 for LR and 0.852 ± 0.023 for HGB, while the class-disparity gap drops from 0.853 to 0.152 for LR and from 0.757 to 0.121 for HGB. The gain is still partial: neither model reaches the nominal 0.90 target on average, and LR Mondrian remains variable on the positive class. APS(allow-empty) stays close to SplitCP and does not remove the class imbalance.

The weighted baseline gives a more mixed picture. For LR it is clearly weaker than Mondrian (0.788 versus 0.876 mean coverage) and still leaves a very large class-disparity gap of 0.659. For HGB it narrows the average coverage gap to -0.046 , close to Mondrian’s marginal performance, but it does so with larger sets and a much worse disparity profile (0.462 versus 0.121). In other words, weighting can reduce average gaps, but it does not give a uniformly clean repair.

The diagnostics support that interpretation. In the without-duration single chronological split, the global calibration–test score Wasserstein distance is already large: 0.173 for LR and 0.177 for HGB. Calibration ESS stays around 1373 for the weighting implementation, so the method is not failing simply because the effective sample size collapses. The harder issue is the combination of clipping distortion and remaining test-side mismatch: the calibration clip fraction is about 0.01, the test-side clip fraction about 0.235, and the raw-versus-clipped calibration TV_{clip} diagnostic about 0.277. Lemma 1 then gives a clipping-only worst-case deviation of about $2 \text{TV}_{\text{clip}} \approx 0.554$. Since this bound is conservative, it should not be used to attribute the observed coverage gaps precisely. Instead, it shows that clipping distortion is one identifiable source of error, while density-ratio estimation error and deeper distributional mismatch remain possible explanations. I therefore treat the weighted baseline as a practical retrospective repair rather than as a theorem-backed deployment solution.

The same ordering appears across $\alpha = 0.05, 0.10$, and 0.20 : vanilla SplitCP and APS under-cover

sharply, Mondrian reduces disparity most cleanly, and weighting provides only a partial, model-dependent reduction in the average coverage gap. What changes with α is the severity, not the direction, of the result.

7.1 Leakage ablation: duration included versus excluded

Table 5: Base-model discrimination under duration ablation.

Dataset	Model	AUC w/o duration	AUC with duration
Original Bank	LR	0.681	0.784
Original Bank	HGB	0.634	0.774
Additional Bank	LR	0.464	0.533
Additional Bank	HGB	0.421	0.661

Including duration increases AUC on both Bank datasets, sometimes by a large margin, but it does not cleanly solve the reliability problem. On Original Bank, LR AUC rises from 0.681 to 0.784 and HGB AUC from 0.634 to 0.774 when duration is included. Even so, the conformal results do not become uniformly reliable: LR Mondrian at $\alpha = 0.10$ still reaches only 0.860 coverage, and HGB Weighted SplitCP changes only slightly from 0.854 without duration (30-run mean) to 0.838 with duration (single seed). The main point is that leakage can improve the ranking layer without fixing the reliability layer.

8 Results III: Original Bank in rolling windows

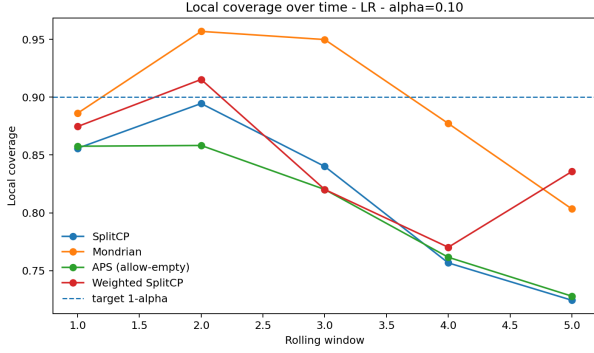
The rolling-window analysis is useful because it shows how coverage changes over time instead of compressing the whole test period into one average. At $\alpha = 0.10$, both the summary table and the trajectories make that variation visible, which is important because the failure is not constant across the test period.

Table 6: Original Bank (`bank-full.csv`), rolling-window summary at $\alpha = 0.10$, without duration.

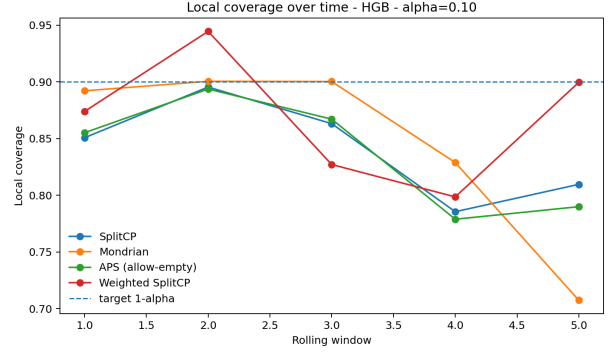
Model	Method	Mean local cov	Min local cov	Max local cov	Mean size	Mean cov. gap
HGB	APS (allow-empty)	0.837	0.779	0.894	1.210	-0.063
HGB	Mondrian	0.846	0.707	0.900	1.592	-0.054
HGB	SplitCP	0.841	0.785	0.895	1.193	-0.059
HGB	Weighted SplitCP	0.869	0.798	0.944	1.294	-0.031
LR	APS (allow-empty)	0.805	0.728	0.858	1.121	-0.095
LR	Mondrian	0.895	0.803	0.957	1.587	-0.005
LR	SplitCP	0.814	0.724	0.894	1.106	-0.086
LR	Weighted SplitCP	0.843	0.770	0.915	1.175	-0.057

Three points stand out. First, vanilla SplitCP remains weak in rolling windows: mean local coverage is 0.814 for LR and 0.841 for HGB. Second, Mondrian again looks like the cleaner repair. For LR it is closest to nominal local coverage on average (0.895 mean local coverage, mean gap -0.005), while the weighted baseline reaches only 0.843 and still carries substantial class disparity in the background diagnostics. Third, on HGB the weighted baseline narrows the average gap more than the unweighted methods, but it still does not restore target coverage and remains less balanced than Mondrian.

Rolling coverage is more informative here than a single global average because the failure is not constant over time. Early and late parts of the test period behave differently, so a global average can hide a sharp local collapse. The summary table makes that visible. For example, at $\alpha = 0.20$ the HGB SplitCP mean local coverage is 0.633, but the minimum window coverage drops much further to 0.290. That gap is a direct sign of time-varying non-exchangeability.



(a) Original Bank, LR, local coverage over time at $\alpha = 0.10$.



(b) Original Bank, HGB, local coverage over time at $\alpha = 0.10$.

Figure 3: Original Bank rolling-window diagnostics at $\alpha = 0.10$. The failure is local and time-varying rather than a single global miss, and the method ranking is strongly model-dependent. For LR, Mondrian stays closest to target on average; for HGB, weighting helps more than it does for LR, but still does not produce a uniformly clean repair.

Looking at the trajectories changes the interpretation. For LR, SplitCP never settles into reliable coverage after the early windows, APS behaves in much the same way, and Mondrian is the only method that stays near target on average. For HGB, weighting helps more than it does for LR, but it still does not provide a uniformly clean fix. The mechanism diagnostics matter here as well. In the 3×10 robustness scan, global score Wasserstein distance and coverage gap are almost perfectly negatively correlated for LR SplitCP ($r = -0.995$), and they remain substantially negative for HGB Mondrian ($r = -0.750$). This is useful because it shows that the issue is not simply that the Bank dataset is difficult. Larger calibration–test score mismatch is directly associated with larger negative coverage gaps.

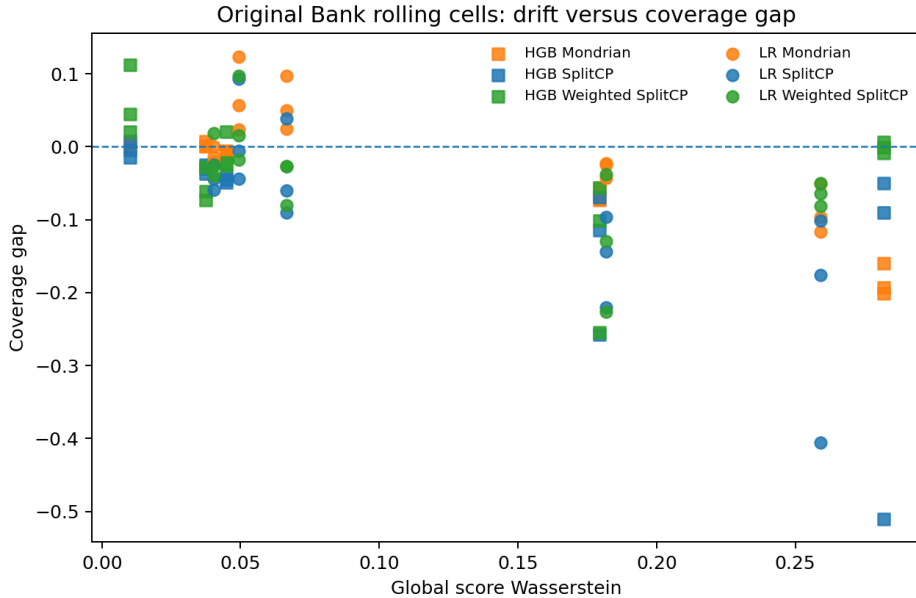


Figure 4: Original Bank rolling cells: global score Wasserstein versus coverage gap. Each point is one (method, window, α) cell, with marker shape indicating model and colour indicating method. Overall Pearson correlation between global score Wasserstein and coverage gap is $r = -0.59$ across all cells.

9 Results IV: Additional Bank as a robustness check

The additional Bank dataset is included as a robustness check rather than a second main experiment. The question is whether the earlier temporal-fragility result survives a change in feature space. It does. If anything, this second dataset makes the role of model dependence even harder to ignore.

Table 7: Additional Bank (`bank-additional-full.csv`), single chronological split, without `duration`, $\alpha = 0.10$.

Model	Method	Cov	Cov(1)	Cov(0)	Avg size	Cov. gap	Dispar.
LR	SplitCP	0.692	0.000	1.000	1.000	-0.208	1.000
LR	Mondrian	0.813	0.394	1.000	1.344	-0.087	0.606
LR	APS (allow-empty)	0.688	0.000	0.994	0.994	-0.213	0.994
LR	Weighted SplitCP	0.953	0.870	0.990	1.593	0.053	0.120
HGB	SplitCP	0.735	0.141	1.000	1.161	-0.165	0.859
HGB	Mondrian	0.837	0.792	0.857	1.703	-0.063	0.065
HGB	APS (allow-empty)	0.730	0.156	0.985	1.183	-0.170	0.829
HGB	Weighted SplitCP	0.946	0.866	0.981	1.583	0.046	0.116

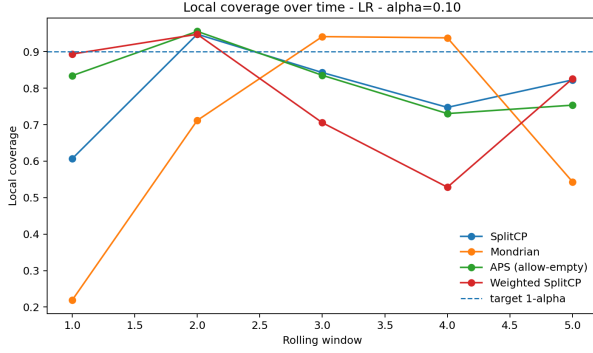
Table 8: Additional Bank (`bank-additional-full.csv`), rolling-window summary at $\alpha = 0.10$, without `duration`.

Model	Method	Mean local cov	Min local cov	Max local cov	Mean size	Mean cov. gap
LR	SplitCP	0.794	0.607	0.948	1.260	-0.106
LR	Mondrian	0.671	0.219	0.942	1.403	-0.229
LR	APS (allow-empty)	0.822	0.731	0.956	1.365	-0.078
LR	Weighted SplitCP	0.959	0.905	1.000	1.716	0.059
HGB	SplitCP	0.835	0.748	0.939	1.181	-0.065
HGB	Mondrian	0.850	0.809	0.883	1.723	-0.050
HGB	APS (allow-empty)	0.828	0.721	0.939	1.187	-0.072
HGB	Weighted SplitCP	0.954	0.889	1.000	1.658	0.054

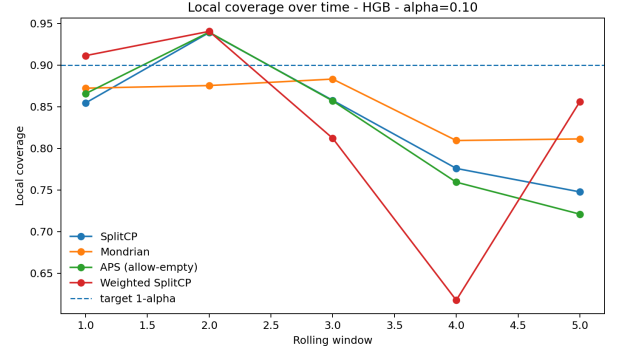
The first important point is not about conformal prediction itself: base-model discrimination on the additional dataset is poor without `duration`, with AUC only 0.464 for LR and 0.421 for HGB. That matters because weak ranking quality already makes reliable and efficient prediction sets harder to obtain. The dataset is therefore best treated as a robustness stress test rather than a clean performance benchmark. Standard conformal procedures are still fragile here. In the single split at $\alpha = 0.10$, LR SplitCP coverage is 0.692 and HGB SplitCP coverage is 0.735. The LR positive-class failure is especially severe, with $\hat{C}_1 = 0.000$ for SplitCP. Mondrian helps HGB substantially but behaves erratically for LR in rolling windows, where its mean local coverage falls to 0.671 and its minimum window coverage falls much further. Weighting is numerically strongest in several cells, but that is not enough to turn this robustness check into a clean leaderboard. The feature space changes, base AUC is weak, and the method ordering becomes even more model-dependent.

The combination of poor AUC and relatively strong weighting performance is still interpretable. AUC measures ranking quality for point prediction, whereas weighting acts on how representative the calibration sample is of the later test block. Those are related, but they are not the same object. The extra macro features do not produce a strong overall ranking model, but they can still help the domain classifier identify calibration cases that look closer to the later test period. In that sense, weighting can remain useful even when the base probabilities are weak: it does not create sharp rankings, but it can shift the threshold toward a more relevant part of covariate space and recover coverage by producing more conservative sets.

The same broad ordering persists across $\alpha = 0.05$, 0.10, and 0.20. WDBC-style near-nominal behaviour does not return; vanilla SplitCP remains fragile; Mondrian helps HGB more than LR; and the weighted baseline remains a model-dependent, assumption-heavy repair rather than a uniformly strongest method. So changing the feature set does not remove the central temporal-fragility result.



(a) Additional Bank, LR, local coverage over time at $\alpha = 0.10$.



(b) Additional Bank, HGB, local coverage over time at $\alpha = 0.10$.

Figure 5: Additional Bank rolling-window diagnostics at $\alpha = 0.10$. The feature-enriched robustness setting does not restore a clean ranking: method ordering remains strongly model-dependent, and any apparent gain from weighting should be read cautiously rather than as a general repair.

The local trajectories make the model dependence easier to see. In LR, class-conditional methods can still fail badly in a late window even when earlier windows look much healthier. In HGB, the weighted baseline often improves on SplitCP, but it still does not turn the problem into a clean solved case. That contrast is exactly why the additional dataset is useful: it shows that the same temporal-stability question can look different depending on how much of the failure comes from weak ranking and how much comes from calibration mismatch.

9.1 Cross-alpha robustness of the main qualitative pattern

The main text focuses on the pre-specified operating point $\alpha = 0.10$ because it offers a useful compromise between conservativeness and failure visibility, but the appendices show that the ordering is not tied to that threshold. Across all three miscoverage levels, WDBC stays close to nominal; vanilla SplitCP and APS under-cover sharply on the Bank datasets; Mondrian often reduces disparity but does not remove temporal fragility; and the weighted baseline remains method- and model-dependent. That consistency across α values matters because it shows the qualitative result is not driven by one chosen threshold.

10 Synthesis table

The synthesis table is selective rather than exhaustive. It includes one benchmark result, one controlled bridge result, one severe temporal failure case, one stronger repair case, and one secondary robustness check.

Table 9: High-level synthesis at $\alpha = 0.10$.

Dataset	Model	Method	Coverage	Cov(1)	Avg size	Interpretation
WDBC baseline	LR	SplitCP	0.893	0.878	0.900	Near-nominal benchmark under approximate exchangeability
WDBC synthetic shift ($\delta = 1.25$)	LR	SplitCP	0.656	0.690	0.736	Controlled bridge experiment: score drift rises and coverage degrades smoothly
Original Bank robustness	LR	SplitCP	0.721 ± 0.027	0.147 ± 0.048	1.109 ± 0.045	Severe temporal failure in all 30 runs; positive-class coverage remains extremely weak
Original Bank robustness	LR	Mondrian	0.876 ± 0.016	0.794 ± 0.127	1.570 ± 0.127	Cleaner repair for class imbalance, but still below nominal on average
Original Bank robustness	HGB	Weighted SplitCP	0.854 ± 0.029	0.538 ± 0.149	1.439 ± 0.160	Partial gap narrowing only; much less clean than Mondrian from a class-disparity perspective
Additional Bank	LR	SplitCP	0.692	0.000	1.000	Feature-enriched robustness check confirms fragility under weak ranking quality

11 Discussion

Five points below summarise the report.

1. **WDBC is a positive control, not the final answer.** Under stratified random splits, the conformal procedures stay close to nominal coverage and show the expected coverage–efficiency

trade-off. This suggests that the implementation is not the main source of the later Bank failures.

2. **The synthetic-shift WDBC experiment identifies the mechanism.** When the WDBC test block is shifted while training and calibration stay fixed, score-Wasserstein distance rises and coverage falls. This supports the interpretation that calibration–test score mismatch is a direct driver of under-coverage.
3. **Chronological Original Bank breaks the naive deployment reading of conformal validity.** Vanilla SplitCP fails in every run of the 3×10 robustness grid at $\alpha = 0.10$. The failure is not only marginal; it is strongly class-asymmetric, especially for the positive class.
4. **Mondrian is the cleaner class-imbalance repair on Original Bank.** Mondrian calibration reduces the positive-class failure and the class-disparity gap more consistently than the weighted baseline. It still does not guarantee nominal coverage under temporal shift, so it should be viewed as a partial repair rather than a complete solution.
5. **Classifier-weighted SplitCP is useful but assumption-heavy.** Weighting can narrow average coverage gaps in some cells, especially for HGB, but it increases set size, often leaves larger class disparity than Mondrian, and uses unlabelled covariates from the future test block. The right interpretation is therefore retrospective batch adjustment, not deployment-ready validity.

The broader lesson is methodological. A conformal prediction result should not be reported only as one marginal coverage number. Under temporal shift and class imbalance, the evaluation needs chronological splits, per-class coverage, rolling-window coverage, set-size summaries, and explicit score-drift diagnostics.

12 Limitations

This study is limited to binary tabular classification, and uses only two base models, one probability-based nonconformity score, and a small set of conformal wrappers. Additional Bank is used only as a robustness check because it has a different feature space and base-model discrimination at the same time. The classifier-weighted baseline mainly targets covariate-shift-like mismatch and can still fail under label shift or concept drift. It also uses unlabelled covariates from the later test block, so the results on the Bank datasets are better read as a retrospective batch deployment-style study than as a strict online forecasting study. Finally, the report evaluates methods under shift more than it develops adaptive methodology. Online or adaptive conformal methods would be natural next comparisons, but were not studied here due to time constraints.

13 Conclusion

The main conclusion is straightforward. Conformal prediction behaves as expected on WDBC when calibration and test data are generated by a more favourable random-split protocol. Once the evaluation becomes chronological, as on Original Bank, vanilla SplitCP under-covers sharply, especially on the positive class, and the robustness grid shows that this is not a one-split accident. Mondrian calibration is the cleanest repair for class imbalance in the main Bank experiment, but it still does not make temporal shift disappear. The classifier-weighted baseline can help in some model–dataset cells, yet its future-test-block information and uneven class-wise behaviour make it a practical batch adjustment rather than a general validity guarantee.

The theoretical lemma in the report has the same restrained interpretation. It bounds the coverage effect of clipping in the weighted baseline, but it does not establish the validity of the whole weighted procedure under temporal shift. The empirical and theoretical parts therefore point to the same message: conformal validity is a calibration claim tied to the relationship between calibration data and deployment data. In time-ordered applications, that relationship has to be tested chronologically and diagnosed explicitly.

A Original Bank: full without-duration single-split table

Tables 10–13 report single-seed runs under the original single-split protocol for full α -grid coverage. At the headline $\alpha = 0.10$ without-duration setting, these single-seed numbers are superseded by the 30-run mean $\pm$ std in Table 4 (main text). The single-seed $\alpha = 0.10$ row is retained here only for cross- α trend reading.

Table 10: Original Bank, single chronological split, without duration, all alpha values.

Reading note: α values are shown in increasing order within each model, and rows are grouped by model/method for easier cross- α comparison.

alpha	Model	Method	Cov	Cov(1)	Cov(0)	Avg size	Cov. gap / Dispar.
0.05	LR	SplitCP	0.810	0.398	1.000	1.251	-0.140 / 0.602
0.05	LR	Mondrian	0.917	0.901	0.924	1.704	-0.033 / 0.023
0.05	LR	APS (allow-empty)	0.841	0.513	0.992	1.378	-0.109 / 0.478
0.05	LR	Weighted SplitCP	0.955	0.857	1.000	1.704	0.005 / 0.143
0.10	LR	SplitCP	0.734	0.159	1.000	1.108	-0.166 / 0.841
0.10	LR	Mondrian	0.890	0.837	0.914	1.574	-0.010 / 0.078
0.10	LR	APS (allow-empty)	0.731	0.217	0.968	1.114	-0.169 / 0.751
0.10	LR	Weighted SplitCP	0.895	0.669	1.000	1.462	-0.005 / 0.331
0.20	LR	SplitCP	0.627	0.000	0.916	0.892	-0.173 / 0.916
0.20	LR	Mondrian	0.810	0.626	0.894	1.281	0.010 / 0.268
0.20	LR	APS (allow-empty)	0.617	0.083	0.864	0.931	-0.183 / 0.780
0.20	LR	Weighted SplitCP	0.806	0.387	1.000	1.243	0.006 / 0.613
0.05	HGB	SplitCP	0.905	0.701	1.000	1.553	-0.045 / 0.299
0.05	HGB	Mondrian	0.923	0.975	0.900	1.826	-0.027 / 0.076
0.05	HGB	APS (allow-empty)	0.891	0.656	0.999	1.599	-0.059 / 0.343
0.05	HGB	Weighted SplitCP	0.983	0.946	1.000	1.882	0.033 / 0.054
0.10	HGB	SplitCP	0.774	0.285	1.000	1.204	-0.126 / 0.715
0.10	HGB	Mondrian	0.881	0.942	0.854	1.699	-0.019 / 0.088
0.10	HGB	APS (allow-empty)	0.774	0.302	0.992	1.224	-0.126 / 0.690
0.10	HGB	Weighted SplitCP	0.982	0.942	1.000	1.875	0.082 / 0.058
0.20	HGB	SplitCP	0.588	0.000	0.860	0.827	-0.212 / 0.860
0.20	HGB	Mondrian	0.797	0.884	0.756	1.490	-0.003 / 0.127
0.20	HGB	APS (allow-empty)	0.646	0.085	0.905	0.967	-0.154 / 0.821
0.20	HGB	Weighted SplitCP	0.849	0.523	1.000	1.374	0.049 / 0.477

B Original Bank: rolling-window summary table

Table 11: Original Bank, rolling-window summary, without duration, all alpha values.

Reading note: α values are shown in increasing order within each model, and rows are grouped by model/method for easier cross- α comparison.

alpha	Model	Method	Mean local cov	Min local cov	Max local cov	Mean size / Mean cov. gap
0.05	HGB	APS (allow-empty)	0.914	0.886	0.938	1.517 / - 0.036
0.05	HGB	Mondrian	0.895	0.749	0.954	1.713 / - 0.055
0.05	HGB	SplitCP	0.911	0.881	0.935	1.476 / - 0.039
0.05	HGB	Weighted SplitCP	0.934	0.894	0.971	1.590 / - 0.016
0.10	HGB	APS (allow-empty)	0.837	0.779	0.894	1.210 / - 0.063
0.10	HGB	Mondrian	0.846	0.707	0.900	1.592 / - 0.054
0.10	HGB	SplitCP	0.841	0.785	0.895	1.193 / - 0.059
0.10	HGB	Weighted SplitCP	0.869	0.798	0.944	1.294 / - 0.031
0.20	HGB	APS (allow-empty)	0.719	0.626	0.791	0.975 / - 0.081
0.20	HGB	Mondrian	0.757	0.640	0.808	1.385 / - 0.043
0.20	HGB	SplitCP	0.633	0.290	0.808	0.799 / - 0.167
0.20	HGB	Weighted SplitCP	0.762	0.546	0.912	1.055 / - 0.038
0.05	LR	APS (allow-empty)	0.892	0.858	0.927	1.406 / - 0.058
0.05	LR	Mondrian	0.928	0.833	0.975	1.725 / - 0.022
0.05	LR	SplitCP	0.879	0.849	0.925	1.304 / - 0.071
0.05	LR	Weighted SplitCP	0.915	0.900	0.932	1.446 / - 0.035
0.10	LR	APS (allow-empty)	0.805	0.728	0.858	1.121 / - 0.095
0.10	LR	Mondrian	0.895	0.803	0.957	1.587 / - 0.005
0.10	LR	SplitCP	0.814	0.724	0.894	1.106 / - 0.086
0.10	LR	Weighted SplitCP	0.843	0.770	0.915	1.175 / - 0.057
0.20	LR	APS (allow-empty)	0.689	0.558	0.768	0.928 / - 0.111
0.20	LR	Mondrian	0.825	0.749	0.924	1.351 / 0.025
0.20	LR	SplitCP	0.689	0.394	0.893	0.878 / - 0.111
0.20	LR	Weighted SplitCP	0.756	0.574	0.897	1.002 / - 0.044

C Additional Bank: full without-duration single-split table

This appendix is included as descriptive support for the robustness-check framing in the main text, not as a second equally weighted leaderboard. Strong single-seed weighted rows here do not overturn the main interpretation because the additional dataset changes the feature space, the base-model discrimination, and the role of weighting at the same time.

Table 12: Additional Bank, single chronological split, without duration, all alpha values.

Reading note: α values are shown in increasing order within each model, and rows are grouped by model/method for easier cross- α comparison.

alpha	Model	Method	Cov	Cov(1)	Cov(0)	Avg size	Cov. gap / Dispar.
0.05	LR	SplitCP	0.756	0.208	1.000	1.147	-0.194 / 0.792
0.05	LR	Mondrian	0.832	0.454	1.000	1.477	-0.118 / 0.547
0.05	LR	APS (allow-empty)	0.760	0.221	1.000	1.169	-0.190 / 0.779
0.05	LR	Weighted SplitCP	0.964	0.882	1.000	1.617	0.014 / 0.118
0.05	HGB	SplitCP	0.817	0.405	1.000	1.486	-0.133 / 0.595
0.05	HGB	Mondrian	0.895	0.887	0.899	1.814	-0.055 / 0.012
0.05	HGB	APS (allow-empty)	0.843	0.495	0.999	1.545	-0.107 / 0.504
0.05	HGB	Weighted SplitCP	0.974	0.917	1.000	1.741	0.024 / 0.083
0.10	LR	SplitCP	0.692	0.000	1.000	1.000	-0.208 / 1.000
0.10	LR	Mondrian	0.813	0.394	1.000	1.344	-0.087 / 0.606
0.10	LR	APS (allow-empty)	0.688	0.000	0.994	0.994	-0.213 / 0.994
0.10	LR	Weighted SplitCP	0.953	0.870	0.990	1.593	0.053 / 0.120
0.10	HGB	SplitCP	0.735	0.141	1.000	1.161	-0.165 / 0.859
0.10	HGB	Mondrian	0.837	0.792	0.857	1.703	-0.063 / 0.065
0.10	HGB	APS (allow-empty)	0.730	0.156	0.985	1.183	-0.170 / 0.829
0.10	HGB	Weighted SplitCP	0.946	0.866	0.981	1.583	0.046 / 0.116
0.20	LR	SplitCP	0.692	0.000	1.000	1.000	-0.108 / 1.000
0.20	LR	Mondrian	0.790	0.319	1.000	1.247	-0.010 / 0.681
0.20	LR	APS (allow-empty)	0.609	0.000	0.881	0.878	-0.191 / 0.881
0.20	LR	Weighted SplitCP	0.931	0.844	0.969	1.550	0.131 / 0.125
0.20	HGB	SplitCP	0.593	0.000	0.857	0.860	-0.207 / 0.857
0.20	HGB	Mondrian	0.770	0.674	0.812	1.564	-0.030 / 0.138
0.20	HGB	APS (allow-empty)	0.635	0.076	0.885	0.936	-0.165 / 0.808
0.20	HGB	Weighted SplitCP	0.887	0.826	0.914	1.501	0.087 / 0.087

D Additional Bank: rolling-window summary table

Table 13: Additional Bank, rolling-window summary, without duration, all alpha values.

Reading note: α values are shown in increasing order within each model, and rows are grouped by model/method for easier cross- α comparison.

alpha	Model	Method	Mean local cov	Min local cov	Max local cov	Mean size / Mean cov. gap
0.05	LR	SplitCP	0.923	0.841	0.989	1.619 / - 0.027
0.05	LR	Mondrian	0.802	0.604	0.966	1.607 / - 0.148
0.05	LR	APS (allow-empty)	0.901	0.843	0.979	1.579 / - 0.049
0.05	LR	Weighted SplitCP	0.973	0.928	1.000	1.777 / 0.023
0.05	HGB	SplitCP	0.917	0.866	0.977	1.577 / - 0.033
0.05	HGB	Mondrian	0.914	0.876	0.942	1.843 / - 0.036
0.05	HGB	APS (allow-empty)	0.915	0.868	0.972	1.573 / - 0.035
0.05	HGB	Weighted SplitCP	0.979	0.946	1.000	1.805 / 0.029
0.10	LR	SplitCP	0.794	0.607	0.948	1.260 / - 0.106
0.10	LR	Mondrian	0.671	0.219	0.942	1.403 / - 0.229
0.10	LR	APS (allow-empty)	0.822	0.731	0.956	1.365 / - 0.078
0.10	LR	Weighted SplitCP	0.959	0.905	1.000	1.716 / 0.059
0.10	HGB	SplitCP	0.835	0.748	0.939	1.181 / - 0.065
0.10	HGB	Mondrian	0.850	0.809	0.883	1.723 / - 0.050
0.10	HGB	APS (allow-empty)	0.828	0.721	0.939	1.187 / - 0.072
0.10	HGB	Weighted SplitCP	0.954	0.889	1.000	1.658 / 0.054
0.20	LR	SplitCP	0.494	0.002	0.843	0.642 / - 0.306
0.20	LR	Mondrian	0.563	0.120	0.928	1.218 / - 0.237
0.20	LR	APS (allow-empty)	0.723	0.612	0.921	1.132 / - 0.077
0.20	LR	Weighted SplitCP	0.929	0.879	0.999	1.639 / 0.129
0.20	HGB	SplitCP	0.645	0.413	0.812	0.820 / - 0.155
0.20	HGB	Mondrian	0.750	0.714	0.791	1.533 / - 0.050
0.20	HGB	APS (allow-empty)	0.718	0.573	0.839	0.946 / - 0.082
0.20	HGB	Weighted SplitCP	0.913	0.804	0.999	1.543 / 0.113

E Protocol note for the exploratory TWM extension

This protocol note applies to the exploratory TWM extension reported in Appendix F. It should not be read as a direct numerical extension of the main rolling summaries in Tables 6 and 8. I produced the TWM experiment with a separate script, and it differs from the core rolling experiment in several ways. The table below lists those differences so that any later comparison is read as qualitative or mechanistic rather than as a same-protocol leaderboard comparison.

Table 14: Protocol differences between the core rolling experiment and the exploratory TWM comparison.

Item	Main rolling experiment	Appendix F exploratory TWM comparison
Rolling construction	Overlapping chronological windows with train/calibration/test fractions 0.50/0.20/0.10 and step size 0.05 (five windows on the Bank datasets).	Fixed 60% training block; the remaining 40% is split into ten sequential chunks paired into five calibration/test windows.
Base-model hyperparameters	Core Bank script defaults; in particular HGB uses learning rate 0.08 and 200 boosting iterations.	Separate exploratory script defaults; HGB uses learning rate 0.05 and 300 boosting iterations, with additional TWM-specific settings.
Weighted baseline definition	Test-point-specific weighted threshold under the main batch, test-block-aware weighting implementation.	Calibration-weighted global threshold in the exploratory script; not numerically identical to the main-script weighted baseline.
Direct comparability to Tables 6 and 8	Yes.	No; comparisons should be read as qualitative unless the same rolling protocol and hyperparameters are re-run.

F Exploratory extension: temporally weighted class-conditional calibration

The main comparison in the report stays with SplitCP, Mondrian SplitCP, and classifier-based Weighted SplitCP. The Bank results nevertheless suggest one natural extension. If temporal drift makes older calibration points less representative of later deployment, and if class imbalance makes that problem class-asymmetric, then class-specific thresholds may benefit from weighting recent calibration points more heavily. That is the motivation for the exploratory temporally weighted Mondrian-style procedure used here.

For calibration tuples (X_i, Y_i, t_i) with time index t_i , the nonconformity score remains

$$s(X_i, Y_i) = 1 - \hat{p}(Y_i | X_i).$$

For a window-level target time t^* , define a normalised time gap

$$\tilde{\Delta}_i(t^*) = \frac{t^* - t_i}{t^* - \min_j t_j},$$

and exponential time-decay weights

$$w_i(t^*) = \exp(-\lambda \tilde{\Delta}_i(t^*)), \quad \lambda \geq 0.$$

When $\lambda = 0$, the weights are constant and the construction reduces to unweighted class-conditional calibration. For each class $y \in \{0, 1\}$, the class-specific weighted empirical distribution is

$$\hat{F}_y(q; t^*) = \frac{\sum_{i: Y_i=y} w_i(t^*) \mathbf{1}\{S_i \leq q\}}{\sum_{i: Y_i=y} w_i(t^*)},$$

and the threshold is

$$\hat{q}_{1-\alpha}^{(y)}(t^*) = \inf\{q : \hat{F}_y(q; t^*) \geq 1 - \alpha\}.$$

The prediction set is then

$$\hat{\Gamma}_\alpha(x; t^*) = \{y \in \{0, 1\} : s(x, y) \leq \hat{q}_{1-\alpha}^{(y)}(t^*)\}.$$

In the implementation used here, t^* is fixed at the rolling-window level instead of being chosen separately for each test point; λ is treated as a pre-specified sensitivity parameter with grid $\{0, 1, 2, 5\}$; and if the class-specific effective sample size under the weights falls below a safety threshold, the procedure falls back to unweighted Mondrian calibration for that class. The design is deliberately practical. It should be read as an operational repair for temporally ordered deployment, not as a claim that finite-sample distribution-free validity has been restored under arbitrary shift.

Proposition 2 (Idealised weak guarantee for class-weighted Mondrian calibration). *Fix a class $y \in \{0, 1\}$. Suppose the class-conditional density ratio*

$$r_y(x) = \frac{dP_{\text{test}}(X \mid Y = y)}{dP_{\text{cal}}(X \mid Y = y)}$$

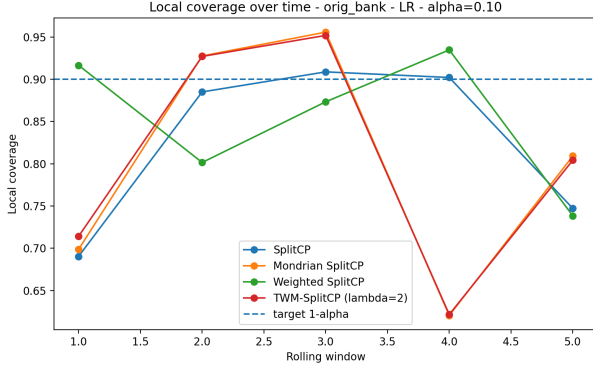
is known, calibration points in class y are weighted proportionally to $r_y(X_i)$, and the weighted class- y calibration scores together with the future class- y test score satisfy the weighted-exchangeability assumptions used in idealised weighted conformal prediction. Then the class-specific weighted quantile $\hat{q}_{1-\alpha}^{(y),w}$ yields

$$\mathbb{P}(Y_{n+1} \in \hat{\Gamma}_\alpha(X_{n+1}) \mid Y_{n+1} = y) \geq 1 - \alpha.$$

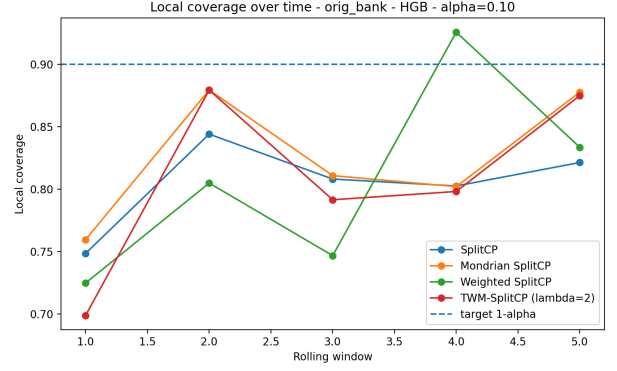
Proof sketch. This is not a new theorem; it is the class-stratified analogue of ideal weighted split conformal calibration. Within a fixed class y , weighting by the known ratio r_y changes the calibration empirical measure so that it matches the class- y test covariate law. Under the corresponding weighted rank argument, the future class- y score has the correct weighted rank among the calibration scores plus itself, so the weighted $(1 - \alpha)$ quantile gives class-conditional coverage at level $1 - \alpha$. The present TWM implementation does *not* satisfy these assumptions exactly: it uses heuristic time-decay weights rather than known density ratios, and it operates with window-level t^* and ESS safeguards. The proposition is therefore only a weak idealisation meant to clarify what sort of object TWM is trying to approximate.

Table 15: Exploratory TWM-SplitCP comparison at $\alpha = 0.10$ (rolling-window summary, without duration). These numbers are not directly comparable to the core rolling summaries in Tables 6 and 8 unless the same rolling protocol and hyperparameters are used.

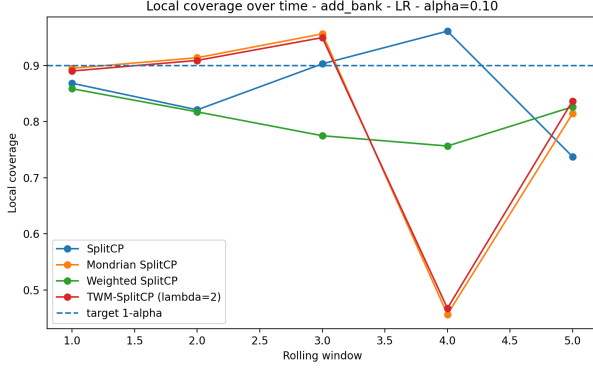
Dataset	Model	Method	Mean local cov	Min local cov	Mean size	Mean cov(1)	Min cov(1)
Original	LR	Mondrian	0.802	0.621	1.580	0.872	0.689
Original	LR	Weighted SplitCP	0.853	0.738	1.238	0.263	0.000
Original	LR	TWM-SplitCP ($\lambda = 2$)	0.804	0.622	1.587	0.874	0.699
Original	HGB	Mondrian	0.826	0.760	1.687	0.847	0.577
Original	HGB	Weighted SplitCP	0.807	0.725	1.201	0.295	0.000
Original	HGB	TWM-SplitCP ($\lambda = 2$)	0.809	0.699	1.666	0.849	0.577
Additional	LR	Mondrian	0.807	0.457	1.446	0.673	0.215
Additional	LR	Weighted SplitCP	0.807	0.757	1.144	0.212	0.000
Additional	LR	TWM-SplitCP ($\lambda = 2$)	0.811	0.468	1.447	0.670	0.177
Additional	HGB	Mondrian	0.923	0.885	1.841	0.892	0.842
Additional	HGB	Weighted SplitCP	0.830	0.637	1.294	0.354	0.000
Additional	HGB	TWM-SplitCP ($\lambda = 2$)	0.919	0.873	1.843	0.893	0.804



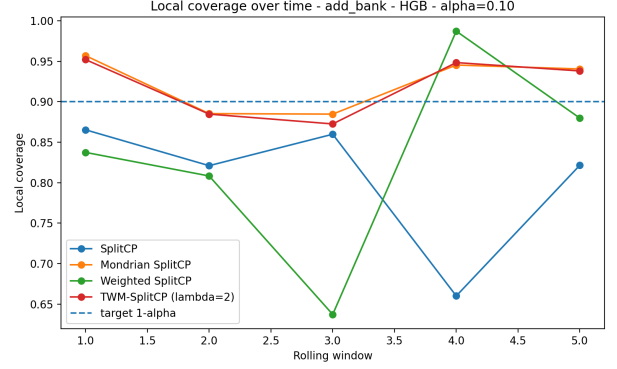
(a) Original Bank, LR, local coverage at $\alpha = 0.10$.



(b) Original Bank, HGB, local coverage at $\alpha = 0.10$.



(c) Additional Bank, LR, local coverage at $\alpha = 0.10$.



(d) Additional Bank, HGB, local coverage at $\alpha = 0.10$.

Figure 6: Exploratory local-coverage comparison for TWM-SplitCP under the exploratory TWM protocol. The extension sometimes improves LR relative to standard Mondrian calibration, especially on the additional dataset, but it does not produce a stable advantage for HGB.

Interpretation. The exploratory evidence is suggestive but limited. On the additional dataset with LR, TWM-SplitCP gives modest gains over unweighted Mondrian, and its class-conditional structure keeps positive-class coverage much higher than vanilla SplitCP or classifier-weighted SplitCP. The improvement is not large enough, or stable enough across both models, to replace the simpler baselines in the main report. In particular, HGB does not show a clear benefit from explicit temporal weighting. I therefore treat TWM-SplitCP as an exploratory extension rather than as a central empirical result.

G Reproducibility note

Table 16: Table–script–parameter mapping for the main empirical artefacts.

Table/Figure	Script	Key settings	Dataset	Notes
Table 4	<code>bank_conformal_experimen ts_patched.py + run_robu stness_scan_patched.py</code>	3×10 grid: <code>train_frac</code> $\in \{0.55, 0.60, 0.65\} \times 10$ seed values; $\alpha = 0.10$; LR/HGB; duration excluded.	Original Bank	Robustness scan; supersedes the earlier single-seed Table 4. The 30-run mean $\pm$ std requires a wrapper loop over <code>train_frac</code> $\in$ $\{0.55, 0.60, 0.65\}$ and 10 seed values. For HGB, the seed changes the base learner. For deterministic LR-based methods, the seed repetition does not induce additional base-model variation and mainly keeps the aggregation protocol aligned across methods.
Table 6, Figure 3	<code>bank_conformal_experimen ts_patched.py</code>	Rolling 0.50/0.20/0.10 with step 0.05 (five windows); LR/HGB; $\alpha \in \{0.05, 0.10, 0.20\}$; single seed; duration excluded.	Original Bank	Core rolling protocol with score-drift and weighting diagnostics enabled.
Table 7, Table 8, Figure 5	<code>bank_conformal_experimen ts_patched.py</code>	Same core protocol as above; duration excluded in the substantive analysis.	Additional Bank	Core Bank protocol; directly comparable within the main text.
Table 15, Figure 6	<code>twm_bank_experiment.py</code>	Fixed 60% training block; five sequential calibration/test window pairs; $\lambda \in \{0, 1, 2, 5\}$ with report value $\lambda = 2$; separate HGB defaults.	Original & Additional Bank	Exploratory protocol; not directly comparable without re-running the same setup.

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